GRAVITON LIBRARY SYSTEM

COMPLETE SOURCE CODE

SINGLE-PAGE WEB APPLICATION

📄 File: index.html
📏 Size: 93,937 characters
📝 Lines: 1667
🛠️ Tech Stack: React 18 · Tailwind CSS · Babel

Generated: 10 Oct 2026 19:21

index.html — GRAVITON LIBRARY SYSTEM

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8"/>
<meta name="viewport" content="width=device-width,initial-scale=1.0"/>
<title>GRAVITON LIBRARY SYSTEM</title>
<link href="https://fonts.googleapis.com/css2?family=Orbitron:wght@400;600;700;900&family=Rajdhani:wght@300;400;500;600;700&display=swap" rel="stylesheet"/>
<script src="https://cdn.tailwindcss.com"></script>
<script src="https://unpkg.com/react@18/umd/react.production.min.js"></script>
<script src="https://unpkg.com/react-dom@18/umd/react-dom.production.min.js"></script>
<script src="https://unpkg.com/@babel/standalone/babel.min.js"></script>
<style>
  *{box-sizing:border-box;margin:0;padding:0}
  :root{
    --cyan:#00FFFF;--purple:#7B2FBE;--orange:#FF6B00;
    --text:#E0F7FA;--success:#00FF88;--danger:#FF3860;
    --bg:#000008;
  }
  html,body,#root{height:100%;width:100%;overflow:hidden}
  body{font-family:'Rajdhani',sans-serif;background:var(--bg);color:var(--text)}
  /* Scrollbar */
  ::-webkit-scrollbar{width:6px;height:6px}
  ::-webkit-scrollbar-track{background:rgba(0,255,255,0.05)}
  ::-webkit-scrollbar-thumb{background:var(--cyan);border-radius:3px}
  /* Fonts */
  .font-orbitron{font-family:'Orbitron',sans-serif}
  .font-rajdhani{font-family:'Rajdhani',sans-serif}
  /* Glow text */
  .glow-cyan{text-shadow:0 0 10px #00FFFF,0 0 20px #00FFFF,0 0 40px #00FFFF}
  .glow-purple{text-shadow:0 0 10px #7B2FBE,0 0 20px #7B2FBE}
  .glow-orange{text-shadow:0 0 10px #FF6B00,0 0 20px #FF6B00}
  .glow-success{text-shadow:0 0 10px #00FF88,0 0 20px #00FF88}
  .glow-danger{text-shadow:0 0 10px #FF3860,0 0 20px #FF3860}
  /* Glass card */
  .glass{
    background:rgba(0,10,30,0.85);
    backdrop-filter:blur(12px);
    -webkit-backdrop-filter:blur(12px);
    border:1px solid rgba(0,255,255,0.3);
    border-radius:12px;
    box-shadow:0 0 20px rgba(0,255,255,0.1),inset 0 0 20px rgba(0,255,255,0.02);
  }
  .glass:hover{
    border-color:rgba(0,255,255,0.7);
    box-shadow:0 0 30px rgba(0,255,255,0.25),inset 0 0 30px rgba(0,255,255,0.05);
    transform:translateY(-6px);
    transition:all 0.3s ease;
  }
  .glass-static{
    background:rgba(0,10,30,0.85);
    backdrop-filter:blur(12px);
    -webkit-backdrop-filter:blur(12px);
    border:1px solid rgba(0,255,255,0.3);
    border-radius:12px;
    box-shadow:0 0 20px rgba(0,255,255,0.1);
  }
  /* Button gradient */
  .btn-primary{
    background:linear-gradient(135deg,var(--cyan),var(--purple));
    color:#000;font-weight:700;
    border:none;border-radius:8px;
    padding:10px 22px;cursor:pointer;
    font-family:'Orbitron',sans-serif;font-size:0.75rem;letter-spacing:1px;
    transition:all 0.3s ease;
```

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      box-shadow:0 0 15px rgba(0,255,255,0.4);
      position:relative;overflow:hidden;
    }
    .btn-primary:hover{
      transform:translateY(-3px);
      box-shadow:0 0 30px rgba(0,255,255,0.7),0 0 60px rgba(123,47,190,0.4);
    }
    .btn-danger{
      background:linear-gradient(135deg,#FF3860,#7B2FBE);
      color:#fff;font-weight:700;
      border:none;border-radius:8px;
      padding:8px 16px;cursor:pointer;
      font-family:'Rajdhani',sans-serif;font-size:0.85rem;
      transition:all 0.3s ease;
      box-shadow:0 0 10px rgba(255,56,96,0.3);
    }
    .btn-danger:hover{transform:translateY(-2px);box-shadow:0 0 20px rgba(255,56,96,0.6)}
    .btn-sm{
      background:linear-gradient(135deg,var(--cyan),var(--purple));
      color:#000;font-weight:700;
      border:none;border-radius:6px;
      padding:6px 14px;cursor:pointer;
      font-family:'Rajdhani',sans-serif;font-size:0.82rem;
      transition:all 0.3s ease;
      box-shadow:0 0 8px rgba(0,255,255,0.3);
    }
    .btn-sm:hover{transform:translateY(-2px);box-shadow:0 0 18px rgba(0,255,255,0.6)}
    .btn-warning{
      background:linear-gradient(135deg,#FF6B00,#7B2FBE);
      color:#fff;font-weight:700;
      border:none;border-radius:6px;
      padding:6px 14px;cursor:pointer;
      font-family:'Rajdhani',sans-serif;font-size:0.82rem;
      transition:all 0.3s ease;
      box-shadow:0 0 8px rgba(255,107,0,0.3);
    }
    .btn-warning:hover{transform:translateY(-2px);box-shadow:0 0 18px rgba(255,107,0,0.6)}
    .btn-success{
      background:linear-gradient(135deg,#00FF88,#00BFFF);
      color:#000;font-weight:700;
      border:none;border-radius:6px;
      padding:6px 14px;cursor:pointer;
      font-family:'Rajdhani',sans-serif;font-size:0.82rem;
      transition:all 0.3s ease;
      box-shadow:0 0 8px rgba(0,255,136,0.3);
    }
    .btn-success:hover{transform:translateY(-2px);box-shadow:0 0 18px rgba(0,255,136,0.6)}
    /* Input styles */
    .input-field{
      background:rgba(0,20,50,0.8);
      border:none;border-bottom:2px solid var(--cyan);
      color:var(--text);padding:10px 12px;
      border-radius:6px 6px 0 0;width:100%;
      font-family:'Rajdhani',sans-serif;font-size:1rem;
      transition:all 0.3s ease;outline:none;
    }
    .input-field:focus{
      border-bottom-color:#7B2FBE;
      box-shadow:0 2px 0 0 #7B2FBE,0 0 15px rgba(123,47,190,0.3);
      background:rgba(0,30,70,0.9);
    }
    .input-field::placeholder{color:rgba(224,247,250,0.3)}
    select.input-field option{background:#000820;color:var(--text)}
    /* Canvas positioning */
    #starCanvas{position:fixed;top:0;left:0;width:100%;height:100%;z-index:0;pointer-events:none}
    .app-layer{position:relative;z-index:1;height:100vh;display:flex;flex-direction:column}
    /* Animations */
    @keyframes floatUp{from{opacity:0;transform:translateY(40px)}to{opacity:1;transform:translateY(0)}}
    @keyframes pulse-border{0%,100%{border-color:rgba(0,255,255,0.3);box-shadow:0 0 20px rgba(0,255,255,0.1)}50%{borde
r-color:rgba(0,255,255,0.8);box-shadow:0 0 40px rgba(0,255,255,0.35)}}
```

```
    @keyframes pulse-orange{0%,100%{box-shadow:0 0 0 0 rgba(255,107,0,0.4)}50%{box-shadow:0 0 0 8px rgba(255,107,0,0)}}
    @keyframes spin-slow{from{transform:rotate(0deg)}to{transform:rotate(360deg)}}
    @keyframes modalIn{from{opacity:0;transform:scale(0.8) translateY(-20px);filter:blur(8px)}to{opacity:1;transform:scale(1) translateY(0);filter:blur(0)}}
    @keyframes toastIn{from{opacity:0;transform:translateX(120%)}to{opacity:1;transform:translateX(0)}}
    @keyframes toastOut{from{opacity:1;transform:translateX(0)}to{opacity:0;transform:translateX(120%)}}
    @keyframes fadeIn{from{opacity:0}to{opacity:1}}
    @keyframes pageFade{from{opacity:0;transform:translateY(15px)}to{opacity:1;transform:translateY(0)}}
    @keyframes ripple{0%{transform:scale(0);opacity:0.7}100%{transform:scale(4);opacity:0}}
    @keyframes waveExpand{0%{transform:scale(0);opacity:0.6}100%{transform:scale(3);opacity:0}}
    .float-up{animation:floatUp 0.6s ease forwards}
    .pulse-border{animation:pulse-border 3s ease-in-out infinite}
    .modal-in{animation:modalIn 0.35s ease forwards}
    .toast-in{animation:toastIn 0.4s ease forwards}
    .page-fade{animation:pageFade 0.4s ease forwards}
    /* Active sidebar item */
    .nav-active{
      background:rgba(0,255,255,0.12);
      border-left:3px solid var(--cyan);
      box-shadow:inset 0 0 20px rgba(0,255,255,0.05);
    }
    .nav-active .nav-icon{color:var(--cyan);filter:drop-shadow(0 0 8px var(--cyan))}
    .nav-active .nav-label{color:var(--cyan)}
    /* Stat card big number */
    .stat-num{font-family:'Orbitron',sans-serif;font-size:2.2rem;font-weight:900}
    /* Table */
    table{border-collapse:collapse;width:100%}
    th{background:rgba(0,255,255,0.08);color:var(--cyan);font-family:'Orbitron',sans-serif;font-size:0.68rem;letter-spacing:1px;padding:10px 14px;text-align:left;cursor:pointer;user-select:none;white-space:nowrap}
    th:hover{background:rgba(0,255,255,0.15)}
    td{padding:10px 14px;border-bottom:1px solid rgba(0,255,255,0.08);font-size:0.92rem;color:var(--text)}
    tr:hover td{background:rgba(0,255,255,0.04)}
    /* Badge */
    .badge{display:inline-flex;align-items:center;padding:2px 10px;border-radius:20px;font-size:0.75rem;font-weight:700;font-family:'Orbitron',sans-serif;letter-spacing:0.5px}
    .badge-success{background:rgba(0,255,136,0.15);color:#00FF88;border:1px solid rgba(0,255,136,0.4)}
    .badge-danger{background:rgba(255,56,96,0.15);color:#FF3860;border:1px solid rgba(255,56,96,0.4)}
    .badge-warning{background:rgba(255,107,0,0.15);color:#FF6B00;border:1px solid rgba(255,107,0,0.4)}
    .badge-info{background:rgba(0,255,255,0.15);color:#00FFFF;border:1px solid rgba(0,255,255,0.4)}
    .badge-grey{background:rgba(150,150,150,0.15);color:#aaa;border:1px solid rgba(150,150,150,0.3)}
    /* Sidebar */
    .sidebar{width:220px;min-width:220px;background:rgba(0,5,20,0.95);border-right:1px solid rgba(0,255,255,0.15);overflow-y:auto;display:flex;flex-direction:column}
    .sidebar-logo{padding:20px 16px;border-bottom:1px solid rgba(0,255,255,0.15)}
    .nav-item{display:flex;align-items:center;gap:12px;padding:13px 18px;cursor:pointer;transition:all 0.25s ease;border-left:3px solid transparent;font-size:0.92rem}
    .nav-item:hover{background:rgba(0,255,255,0.07);border-left-color:rgba(0,255,255,0.3)}
    .nav-icon{font-size:1.1rem;opacity:0.7;min-width:20px;text-align:center}
    /* Mobile bottom bar */
    .bottom-bar{display:none;position:fixed;bottom:0;left:0;right:0;background:rgba(0,5,20,0.97);border-top:1px solid rgba(0,255,255,0.2);z-index:100;padding:6px 0}
    @media(max-width:768px){
      .sidebar{display:none}
      .bottom-bar{display:flex;justify-content:space-around}
      .main-area{margin-bottom:64px}
      .stat-num{font-size:1.6rem}
    }
    /* Error */
    .field-error{color:#FF3860;font-size:0.8rem;margin-top:4px;font-family:'Rajdhani',sans-serif;text-shadow:0 0 8px rgba(255,56,96,0.5)}
    /* Toggle */
    .toggle-track{width:48px;height:26px;border-radius:13px;background:rgba(100,100,100,0.4);position:relative;cursor:pointer;transition:background 0.3s}
    .toggle-track.on{background:rgba(0,255,255,0.3);box-shadow:0 0 12px rgba(0,255,255,0.4)}
    .toggle-thumb{position:absolute;top:3px;left:3px;width:20px;height:20px;border-radius:50%;background:#888;transition:all 0.3s}
    .toggle-track.on .toggle-thumb{left:25px;background:var(--cyan);box-shadow:0 0 8px var(--cyan)}
    /* Glow number badge */
    .pulse-badge{animation:pulse-orange 2s ease-in-out infinite}
    /* Empty state */
```

```
    .empty-state{text-align:center;padding:60px 20px;color:rgba(0,255,255,0.4);font-family:'Orbitron',sans-serif;font-size:0.85rem;letter-spacing:2px}
    .empty-icon{font-size:3rem;margin-bottom:12px;opacity:0.4}
  </style>
  </head>
  <body>
  <canvas id="starCanvas"></canvas>
  <div id="root"></div>
  <script type="text/babel">
  const {useState,useEffect,useRef,useCallback,useMemo}=React;

  // â•â•â•
```

```
      let waves=[];
      let lastWave=0;
      let frame=0;
      function addWave(){waves.push({r:0,opacity:0.6,born:frame})}
      function draw(){
        frame++;
        ctx.clearRect(0,0,W,H);
        // Gravitational wave rings
        if(frame-lastWave>180){lastWave=frame;addWave()}
        waves=waves.filter(w=>w.opacity>0.01);
        waves.forEach(w=>{
          w.r+=1.8;w.opacity*=0.985;
          ctx.beginPath();
          ctx.arc(W/2,H/2,w.r,0,Math.PI*2);
          ctx.strokeStyle=`rgba(0,255,255,${w.opacity*0.5})`;
          ctx.lineWidth=1.5;ctx.stroke();
          // second ring
          if(w.r>30){
            ctx.beginPath();
            ctx.arc(W/2,H/2,w.r-25,0,Math.PI*2);
            ctx.strokeStyle=`rgba(123,47,190,${w.opacity*0.3})`;
            ctx.lineWidth=1;ctx.stroke();
          }
        });
        // Stars
        stars.forEach(s=>{
          s.angle+=s.speed*0.005;
          s.x+=Math.cos(s.angle)*s.speed;
          s.y+=Math.sin(s.angle)*s.speed;
          if(s.x<0)s.x=W;if(s.x>W)s.x=0;
          if(s.y<0)s.y=H;if(s.y>H)s.y=0;
          // Gravitational lensing warp near center
          const dx=s.x-W/2,dy=s.y-H/2;
          const dist=Math.sqrt(dx*dx+dy*dy);
          const lensR=200;
          let rx=s.x,ry=s.y;
          if(dist<lensR&&dist>5){
            const f=(1-(dist/lensR))*0.15;
            rx=s.x+dx*f;ry=s.y+dy*f;
          }
          s.twinkle+=0.04;
          const twinkle=0.5+0.5*Math.sin(s.twinkle);
          const alpha=0.4+0.6*twinkle;
          ctx.beginPath();
          ctx.arc(rx,ry,s.size,0,Math.PI*2);
          ctx.fillStyle=`rgba(${twinkle>0.7?'0,255,255':'200,220,255'},${alpha})`;
          ctx.fill();
        });
        requestAnimationFrame(draw);
      }
      draw();
      const onResize=()=>{W=window.innerWidth;H=window.innerHeight;canvas.width=W;canvas.height=H};
      window.addEventListener('resize',onResize);
      return()=>window.removeEventListener('resize',onResize);
    },[]);
    return null;
  }

  // â•â•â•
```

```
          borderColor:t.type==='success'?'rgba(0,255,136,0.5)':t.type==='error'?'rgba(255,56,96,0.5)':'rgba(255,107,0,0.5)',
          boxShadow:`0 0 20px ${t.type==='success'?'rgba(0,255,136,0.3)':t.type==='error'?'rgba(255,56,96,0.3)':'rgba(255,107,0,0.3)'}`,
          display:'flex',alignItems:'center',gap:12,cursor:'pointer',
        }} onClick={()=>removeToast(t.id)}>
          <span style={{fontSize:'1.3rem'}}>{t.type==='success'?'âœ…':t.type==='error'?'âŒ':'âš ï¸'}</span>
          <span style={{fontFamily:'Rajdhani',fontSize:'0.95rem',color:t.type==='success'?'#00FF88':t.type==='error'?'#FF3860':'#FF6B00',fontWeight:600}}>{t.message}</span>
        </div>
      ))}
    </div>
  );
}
function useToast(){
  const [toasts,setToasts]=useState([]);
  const addToast=useCallback((message,type='success')=>{
    const id=Date.now();
    setToasts(p=>[...p,{id,message,type}]);
    setTimeout(()=>setToasts(p=>p.filter(t=>t.id!==id)),3000);
  },[]);
  const removeToast=useCallback((id)=>setToasts(p=>p.filter(t=>t.id!==id)),[]);
  return{toasts,addToast,removeToast};
}

// â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•
// MODAL
// â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•
function Modal({open,onClose,title,children,width='500px'}){
  if(!open)return null;
  return(
    <div style={{position:'fixed',inset:0,zIndex:1000,display:'flex',alignItems:'center',justifyContent:'center',padding:'20px'}}
      onClick={e=>{if(e.target===e.currentTarget)onClose()}}>
      <div style={{position:'absolute',inset:0,background:'rgba(0,0,8,0.75)',backdropFilter:'blur(4px)'}}/>
      <div className="glass-static modal-in" style={{position:'relative',width:'100%',maxWidth:width,maxHeight:'90vh',overflowY:'auto',padding:'28px'}}>
        <div style={{display:'flex',justifyContent:'space-between',alignItems:'center',marginBottom:20}}>
          <h2 className="font-orbitron glow-cyan" style={{fontSize:'1rem',letterSpacing:2,color:'var(--cyan)'}}>{title}</h2>
          <button onClick={onClose} style={{background:'none',border:'none',color:'rgba(0,255,255,0.6)',fontSize:'1.4rem',cursor:'pointer',lineHeight:1}}>âœ•</button>
        </div>
        {children}
      </div>
    </div>
  );
}

// â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•
// SORT TABLE HOOK
// â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•
function useSortedData(data,defaultKey=''){
  const [sortKey,setSortKey]=useState(defaultKey);
  const [sortDir,setSortDir]=useState('asc');
  const sort=(key)=>{if(sortKey===key)setSortDir(d=>d==='asc'?'desc':'asc');else{setSortKey(key);setSortDir('asc')}};
  const sorted=useMemo(()=>{
    if(!sortKey)return data;
    return [...data].sort((a,b)=>{
      const av=a[sortKey]??'',bv=b[sortKey]??'';
      const cmp=typeof av==='number'?av-bv:String(av).localeCompare(String(bv));
      return sortDir==='asc'?cmp:-cmp;
    });
  },[data,sortKey,sortDir]);
  const SortIcon=({k})=><span style={{marginLeft:4,opacity:sortKey===k?1:0.3,fontSize:'0.7rem'}}>{sortKey===k&&sortDir==='desc'?'â–¼':'â–²'}</span>;
```

```
    return{sorted,sort,sortKey,sortDir,SortIcon};
  }

  // â•Ââ•Ââ•Ââ
```

```
            </div>
            {error&&<p style={{color:'#FF3860',textShadow:'0 0 8px rgba(255,56,96,0.6)',fontFamily:'Rajdhani',fontSize:'0.9rem',textAlign:'center'}}>{error}</p>}
            <button type="submit" className="btn-primary" style={{width:'100%',padding:'13px',fontSize:'0.9rem',position:'relative',overflow:'hidden'}} onClick={handleRipple}>
              {ripple&&<span style={{position:'absolute',left:ripple.x,top:ripple.y,width:10,height:10,borderRadius:'50%',background:'rgba(255,255,255,0.4)',transform:'translate(-50%,-50%)',animation:'ripple 0.7s ease-out forwards',pointerEvents:'none'}}/>}
              INITIALIZE ACCESS
            </button>
          </form>
          <div style={{marginTop:28,display:'flex',flexDirection:'column',gap:10}}>
            <p style={{textAlign:'center',color:'rgba(0,255,255,0.4)',fontSize:'0.75rem',fontFamily:'Orbitron',letterSpacing:1}}>QUICK ACCESS</p>
            {presets.map(pr=>(
              <button key={pr.u} onClick={()=>quickLogin(pr.u,pr.p)} style={{
                background:'rgba(0,255,255,0.05)',border:'1px solid rgba(0,255,255,0.2)',borderRadius:8,
                color:'var(--text)',padding:'10px 14px',cursor:'pointer',
                display:'flex',justifyContent:'space-between',alignItems:'center',
                fontFamily:'Rajdhani',fontSize:'0.92rem',transition:'all 0.25s',
              }} onMouseOver={e=>e.currentTarget.style.background='rgba(0,255,255,0.1)'}
                 onMouseOut={e=>e.currentTarget.style.background='rgba(0,255,255,0.05)'}>
                <span>{pr.label}</span>
                <span className="badge badge-info" style={{fontSize:'0.65rem'}}>{pr.badge}</span>
              </button>
            ))}
          </div>
        </div>
        <div style={{position:'fixed',bottom:14,left:18,color:'rgba(0,255,255,0.4)',fontSize:'0.7rem',fontFamily:'Orbitron',letterSpacing:1}}>
          â˜… 300 PARTICLES
        </div>
      </div>
    );
  }

// â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•
// HEADER
// â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•
function Header({user,onLogout}){
  return(
    <div style={{height:60,background:'rgba(0,5,20,0.95)',borderBottom:'1px solid rgba(0,255,255,0.2)',display:'flex',alignItems:'center',justifyContent:'space-between',padding:'0 24px',flexShrink:0,zIndex:10}}>
      <div style={{display:'flex',alignItems:'center',gap:12}}>
        <span style={{fontSize:'1.3rem'}}>âš›ï¸</span>
        <span className="font-orbitron glow-cyan" style={{fontSize:'0.75rem',color:'var(--cyan)',letterSpacing:2}}>GRAVITON LIBRARY</span>
      </div>
      <div style={{display:'flex',alignItems:'center',gap:20}}>
        <Clock/>
        <div style={{textAlign:'right'}}>
          <p style={{fontFamily:'Orbitron',fontSize:'0.68rem',color:'rgba(0,255,255,0.6)',letterSpacing:1}}>{user.role.toUpperCase()}</p>
          <p style={{fontFamily:'Rajdhani',fontSize:'0.95rem',fontWeight:600}}>{user.name}</p>
        </div>
        <button onClick={onLogout} className="btn-danger" style={{padding:'6px 14px',fontSize:'0.78rem'}}>LOGOUT</button>
      </div>
    </div>
  );
}

// â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•
// STAT CARD
// â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•â•
function StatCard({icon,label,value,color='var(--cyan)',pulse=false}){
  const isOverZero=value>0&&pulse;
```

```
    return(
      <div className="glass" style={{padding:'20px 24px',display:'flex',flexDirection:'column',gap:8,cursor:'defaul
t'}}>
        <div style={{display:'flex',justifyContent:'space-between',alignItems:'flex-start'}}>
          <span style={{fontSize:'1.6rem'}}>{icon}</span>
          {isOverZero&&<span className="badge badge-warning pulse-badge" style={{fontSize:'0.65rem'}}>{value}</span>}
        </div>
        <p className="stat-num" style={{color,textShadow:`0 0 20px ${color}`}}>{value}</p>
        <p style={{color:'rgba(224,247,250,0.6)',fontFamily:'Rajdhani',fontSize:'0.82rem',letterSpacing:1}}>{label}</p
>
      </div>
    );
  }

  // â•â•â•
```

```
557
558 function LibDashHome({state}){
559   const {books,loans,members,fines}=state;
560   const activeLoans=loans.filter(l=>l.status==='active');
561   const overdue=activeLoans.filter(l=>new Date(l.dueDate)<new Date());
562   const finesCollected=fines.filter(f=>f.paid).reduce((a,b)=>a+b.amount,0);
563   return(
564     <div>
565       <h2 className="font-orbitron glow-cyan" style={{fontSize:'1.1rem',letterSpacing:3,color:'var(--cyan)',marginBo
ttom:24}}>MISSION CONTROL</h2>
566       <div style={{display:'grid',gridTemplateColumns:'repeat(auto-fill,minmax(180px,1fr))',gap:16}}>
567         <StatCard icon="ðŸ“š" label="TOTAL BOOKS" value={books.reduce((a,b)=>a+b.totalCopies,0)} color="var(--cya
n)"/>
568         <StatCard icon="âœ…" label="AVAILABLE" value={books.reduce((a,b)=>a+b.availableCopies,0)} color="var(--succe
ss)"/>
569         <StatCard icon="ðŸ“¤" label="ACTIVE LOANS" value={activeLoans.length} color="var(--cyan)"/>
570         <StatCard icon="âš ï¸" label="OVERDUE" value={overdue.length} color="var(--orange)" pulse/>
571         <StatCard icon="ðŸ‘¥" label="TOTAL MEMBERS" value={members.length} color="var(--purple)"/>
572         <StatCard icon="ðŸ’°" label="FINES (â‚¹)" value={finesCollected} color="var(--success)"/>
573       </div>
574     </div>
575   );
576 }
577
578 // â”€â”€â”€ MANAGE BOOKS â”€â”€â”€â”€
579 function ManageBooks({state,dispatch,addToast}){
580   const [search,setSearch]=useState('');
581   const [modal,setModal]=useState(false);
582   const [editing,setEditing]=useState(null);
583   const [form,setForm]=useState({isbn:'',title:'',author:'',category:'',totalCopies:1});
584   const [errors,setErrors]=useState({});
585   const filtered=state.books.filter(b=>
586     b.title.toLowerCase().includes(search.toLowerCase())||
587     b.author.toLowerCase().includes(search.toLowerCase())||
588     b.isbn.includes(search)||
589     b.category.toLowerCase().includes(search.toLowerCase())
590   );
591   const {sorted,sort,SortIcon}=useSortedData(filtered,'title');
592   const openAdd=()=>{setEditing(null);setForm({isbn:'',title:'',author:'',category:'',totalCopies:1});setErrors({});
setModal(true)};
593   const openEdit=(b)=>{setEditing(b.isbn);setForm({isbn:b.isbn,title:b.title,author:b.author,category:b.category,tot
alCopies:b.totalCopies});setErrors({});setModal(true)};
594   const validate=()=>{
595     const e={};
596     if(!form.isbn.trim())e.isbn='ISBN is required';
597     if(!form.title.trim())e.title='Title is required';
598     if(!form.author.trim())e.author='Author is required';
599     if(!form.category.trim())e.category='Category is required';
600     if(!form.totalCopies||form.totalCopies<1)e.totalCopies='At least 1 copy required';
601     if(!editing&&state.books.find(b=>b.isbn===form.isbn.trim()))e.isbn='ISBN already exists';
602     setErrors(e);return Object.keys(e).length===0;
603   };
604   const handleSave=()=>{
605     if(!validate())return;
606     if(editing){
607       dispatch({type:'EDIT_BOOK',payload:{...form,isbn:editing,totalCopies:parseInt(form.totalCopies)}});
608       addToast('Book updated successfully','success');
609     }else{
610       dispatch({type:'ADD_BOOK',payload:{...form,totalCopies:parseInt(form.totalCopies),availableCopies:parseInt(for
m.totalCopies)}});
611       addToast('Book added to the system','success');
612     }
613     setModal(false);
614   };
615   const handleDelete=(isbn)=>{
616     const hasLoans=state.loans.find(l=>l.bookIsbn===isbn&&l.status==='active');
617     if(hasLoans){addToast('Cannot delete â€” book has active loans','error');return}
618     dispatch({type:'DELETE_BOOK',payload:isbn});
619     addToast('Book removed from system','success');
620   };
```

```
    return(
      <div>
        <div style={{display:'flex',justifyContent:'space-between',alignItems:'center',marginBottom:20,gap:12,flexWrap:'wrap'}}>
          <h2 className="font-orbitron glow-cyan" style={{fontSize:'1rem',letterSpacing:3,color:'var(--cyan)'}}>BOOK CATALOGUE</h2>
          <div style={{display:'flex',gap:10,alignItems:'center'}}>
            <input className="input-field" value={search} onChange={e=>setSearch(e.target.value)} placeholder="ðŸ” Search books..." style={{width:220}}/>
            <button className="btn-primary" onClick={openAdd}>+ ADD BOOK</button>
          </div>
        </div>
        <div className="glass-static" style={{overflowX:'auto'}}>
          {sorted.length===0?<div className="empty-state"><div className="empty-icon">ðŸ“š</div>NO BOOKS FOUND</div>:(
            <table>
              <thead><tr>
                {[['isbn','ISBN'],['title','TITLE'],['author','AUTHOR'],['category','CATEGORY'],['totalCopies','TOTAL'],['availableCopies','AVAILABLE']].map(([k,l])=>(
                  <th key={k} onClick={()=>sort(k)}>{l}<SortIcon k={k}/></th>
                ))}
                <th>ACTIONS</th>
              </tr></thead>
              <tbody>{sorted.map(b=>(
                <tr key={b.isbn}>
                  <td><span className="badge badge-info">{b.isbn}</span></td>
                  <td style={{fontWeight:600}}>{b.title}</td>
                  <td>{b.author}</td>
                  <td><span className="badge badge-grey">{b.category}</span></td>
                  <td style={{textAlign:'center'}}>{b.totalCopies}</td>
                  <td style={{textAlign:'center'}}>
                    <span className={`badge ${b.availableCopies>0?'badge-success':'badge-danger'}`}>{b.availableCopies}</span>
                  </td>
                  <td style={{display:'flex',gap:6}}>
                    <button className="btn-sm" onClick={()=>openEdit(b)}>EDIT</button>
                    <button className="btn-danger" style={{fontSize:'0.78rem',padding:'5px 12px'}} onClick={()=>handleDelete(b.isbn)}>DEL</button>
                  </td>
                </tr>
              ))}</tbody>
            </table>
          )}
        </div>
        <Modal open={modal} onClose={()=>setModal(false)} title={editing?'EDIT BOOK':'ADD NEW BOOK'}>
          {[['isbn','ISBN'],['title','TITLE'],['author','AUTHOR'],['category','CATEGORY']].map(([k,l])=>(
            <div key={k} style={{marginBottom:16}}>
              <label style={{display:'block',fontSize:'0.72rem',color:'rgba(0,255,255,0.6)',fontFamily:'Orbitron',letterSpacing:1,marginBottom:6}}>{l}</label>
              <input className="input-field" value={form[k]} onChange={e=>setForm(p=>({...p,[k]:e.target.value}))} disabled={!!editing&&k==='isbn'}/>
              {errors[k]&&<p className="field-error">{errors[k]}</p>}
            </div>
          ))}
          <div style={{marginBottom:20}}>
            <label style={{display:'block',fontSize:'0.72rem',color:'rgba(0,255,255,0.6)',fontFamily:'Orbitron',letterSpacing:1,marginBottom:6}}>NUMBER OF COPIES</label>
            <input className="input-field" type="number" min="1" value={form.totalCopies} onChange={e=>setForm(p=>({...p,totalCopies:e.target.value}))}/>
            {errors.totalCopies&&<p className="field-error">{errors.totalCopies}</p>}
          </div>
          <button className="btn-primary" style={{width:'100%'}} onClick={handleSave}>{editing?'UPDATE BOOK':'ADD BOOK'}</button>
        </Modal>
      </div>
    );
  }

  // â”€â”€â”€ ISSUE BOOK â”€â”€â”€â”€
  function IssueBook({state,dispatch,addToast}){
    const [memberId,setMemberId]=useState('');
    const [isbn,setIsbn]=useState('');
```

```
    const [errors,setErrors]=useState({});
    const members=state.members;
    const availBooks=state.books.filter(b=>b.availableCopies>0);
    const handleIssue=()=>{
      const e={};
      if(!memberId)e.member='Select a member';
      if(!isbn)e.isbn='Select a book';
      setErrors(e);
      if(Object.keys(e).length)return;
      const member=members.find(m=>m.id===memberId);
      const book=state.books.find(b=>b.isbn===isbn);
      const memberLoans=state.loans.filter(l=>l.memberId===memberId&&l.status==='active');
      if(memberLoans.length>=member.maxBooks){addToast(`${member.name} has reached max borrow limit (${member.maxBook
s})!`,'error');return}
      const issueDate=today();
      const dueDate=addDays(issueDate,15);
      dispatch({type:'ISSUE_BOOK',payload:{memberId,bookIsbn:isbn,issueDate,dueDate,memberName:member.name,bookTitle:b
ook.title}});
      addToast(`"${book.title}" issued to ${member.name}`,'success');
      setMemberId('');setIsbn('');
    };
    return(
      <div>
        <h2 className="font-orbitron glow-cyan" style={{fontSize:'1rem',letterSpacing:3,color:'var(--cyan)',marginBott
om:24}}>ISSUE BOOK</h2>
        <div className="glass-static" style={{padding:28,maxWidth:500}}>
          <div style={{marginBottom:20}}>
            <label style={{display:'block',fontSize:'0.72rem',color:'rgba(0,255,255,0.6)',fontFamily:'Orbitron',letter
Spacing:1,marginBottom:8}}>SELECT MEMBER</label>
            <select className="input-field" value={memberId} onChange={e=>setMemberId(e.target.value)}>
              <option value="">-- Choose Member --</option>
              {members.map(m=><option key={m.id} value={m.id}>{m.name} ({m.membershipId}) â€” {m.memberType}</option
>)}
            </select>
            {errors.member&&<p className="field-error">{errors.member}</p>}
          </div>
          <div style={{marginBottom:24}}>
            <label style={{display:'block',fontSize:'0.72rem',color:'rgba(0,255,255,0.6)',fontFamily:'Orbitron',letter
Spacing:1,marginBottom:8}}>SELECT BOOK (AVAILABLE ONLY)</label>
            <select className="input-field" value={isbn} onChange={e=>setIsbn(e.target.value)}>
              <option value="">-- Choose Book --</option>
              {availBooks.map(b=><option key={b.isbn} value={b.isbn}>{b.title} â€” {b.availableCopies} avail.</option
>)}
            </select>
            {errors.isbn&&<p className="field-error">{errors.isbn}</p>}
          </div>
          {memberId&&isbn&&(()=>{
            const m=members.find(x=>x.id===memberId);
            const b=state.books.find(x=>x.isbn===isbn);
            return(
              <div className="glass" style={{padding:14,marginBottom:20,cursor:'default'}}>
                <p style={{fontSize:'0.8rem',color:'rgba(0,255,255,0.6)',fontFamily:'Orbitron',letterSpacing:1,marginB
ottom:8}}>ISSUE PREVIEW</p>
                <p style={{fontFamily:'Rajdhani',fontSize:'0.95rem'}}><b>{b?.title}</b> â†’ <b>{m?.name}</b></p>
                <p style={{fontFamily:'Rajdhani',fontSize:'0.85rem',color:'rgba(224,247,250,0.6)'}}>Issue: {formatDate
(today())} | Due: {formatDate(addDays(today(),15))}</p>
              </div>
            );
          })()}
          <button className="btn-primary" style={{width:'100%'}} onClick={handleIssue}>ISSUE BOOK</button>
        </div>
      </div>
    );
  }

  // â”€â”€â”€ RETURN BOOK â”€â”€â”€
  function ReturnBook({state,dispatch,addToast}){
    const [search,setSearch]=useState('');
    const activeLoans=state.loans.filter(l=>l.status==='active');
    const filtered=activeLoans.filter(l=>
      l.memberName.toLowerCase().includes(search.toLowerCase())||
```

```
      l.bookTitle.toLowerCase().includes(search.toLowerCase())||
      l.id.toLowerCase().includes(search.toLowerCase())
    );
    const {sorted,sort,SortIcon}=useSortedData(filtered,'memberName');
    const handleReturn=(loan)=>{
      const today_=today();
      const overdueDays=daysBetween(loan.dueDate,today_);
      const fineAmount=overdueDays>0?overdueDays*state.settings.fineRate:0;
      dispatch({type:'RETURN_BOOK',payload:{loanId:loan.id,returnDate:today_,fineAmount}});
      // check reservation
      const res=state.reservations.find(r=>r.bookIsbn===loan.bookIsbn&&r.status==='pending');
      if(res){addToast(`ðŸ“Œ Reservation pending for "${loan.bookTitle}" â€” notify ${res.memberName}`,'warning')}
      if(fineAmount>0){addToast(`"${loan.bookTitle}" returned. Fine: â‚¹${fineAmount} (${overdueDays} days overdue)`,'warning')}
      else{addToast(`"${loan.bookTitle}" returned successfully`,'success')}
    };
    return(
      <div>
        <div style={{display:'flex',justifyContent:'space-between',alignItems:'center',marginBottom:20,flexWrap:'wrap',gap:12}}>
          <h2 className="font-orbitron glow-cyan" style={{fontSize:'1rem',letterSpacing:3,color:'var(--cyan)'}}>RETURN BOOK</h2>
          <input className="input-field" value={search} onChange={e=>setSearch(e.target.value)} placeholder="ðŸ”Ž Search loans..." style={{width:240}}/>
        </div>
        <div className="glass-static" style={{overflowX:'auto'}}>
          {sorted.length===0?<div className="empty-state"><div className="empty-icon">ðŸ“¥</div>NO ACTIVE LOANS</div>:(
            <table>
              <thead><tr>
                {[['id','LOAN ID'],['memberName','MEMBER'],['bookTitle','BOOK'],['issueDate','ISSUED'],['dueDate','DUE DATE']].map(([k,l])=>(
                  <th key={k} onClick={()=>sort(k)}>{l}<SortIcon k={k}/></th>
                ))}
                <th>STATUS</th><th>FINE (â‚¹)</th><th>ACTION</th>
              </tr></thead>
              <tbody>{sorted.map(l=>{
                const today_=today();
                const overdueDays=daysBetween(l.dueDate,today_);
                const isOverdue=overdueDays>0;
                const fine=isOverdue?overdueDays*state.settings.fineRate:0;
                return(
                  <tr key={l.id}>
                    <td><span className="badge badge-info">{l.id}</span></td>
                    <td>{l.memberName}</td>
                    <td style={{fontWeight:600}}>{l.bookTitle}</td>
                    <td>{formatDate(l.issueDate)}</td>
                    <td style={{color:isOverdue?'#FF3860':'var(--text)'}}>{formatDate(l.dueDate)}</td>
                    <td>{isOverdue?<span className="badge badge-danger">OVERDUE {overdueDays}d</span>:<span className="badge badge-success">ACTIVE</span>}</td>
                    <td style={{color:fine>0?'#FF3860':'var(--success)',fontWeight:700}}>â‚¹{fine}</td>
                    <td><button className="btn-success" onClick={()=>handleReturn(l)}>RETURN</button></td>
                  </tr>
                );
              })}</tbody>
            </table>
          )}
        </div>
      </div>
    );
  }

  // â”€â”€â”€ MANAGE MEMBERS â”€â”€â”€
  function ManageMembers({state,dispatch,addToast}){
    const [search,setSearch]=useState('');
    const [modal,setModal]=useState(false);
    const [form,setForm]=useState({name:'',username:'',password:'',membershipId:'',memberType:'Student',maxBooks:3,email:''});
    const [errors,setErrors]=useState({});
    const filtered=state.members.filter(m=>
```

```
      m.name.toLowerCase().includes(search.toLowerCase())||
      m.membershipId?.toLowerCase().includes(search.toLowerCase())||
      m.username.toLowerCase().includes(search.toLowerCase())
    );
    const {sorted,sort,SortIcon}=useSortedData(filtered,'name');
    const validate=()=>{
      const e={};
      if(!form.name.trim())e.name='Name is required';
      if(!form.username.trim())e.username='Username is required';
      if(state.members.find(m=>m.username===form.username.trim())||INITIAL_USERS.find(u=>u.username===form.username.tr
im()&&u.role!=='member'))e.username='Username already taken';
      if(!form.password||form.password.length<4)e.password='Min 4 characters';
      if(!form.membershipId.trim())e.membershipId='Membership ID required';
      if(state.members.find(m=>m.membershipId===form.membershipId.trim()))e.membershipId='ID already exists';
      setErrors(e);return Object.keys(e).length===0;
    };
    const handleSave=()=>{
      if(!validate())return;
      const id='u'+Date.now();
      dispatch({type:'ADD_MEMBER',payload:{id,role:'member',...form,maxBooks:parseInt(form.maxBooks)}});
      addToast(`Member ${form.name} registered`,'success');
      setModal(false);setForm({name:'',username:'',password:'',membershipId:'',memberType:'Student',maxBooks:3,emai
l:''});
    };
    const handleDelete=(id,name)=>{
      const hasLoans=state.loans.find(l=>l.memberId===id&&l.status==='active');
      if(hasLoans){addToast(`${name} has active loans â€" cannot delete`,'error');return}
      dispatch({type:'DELETE_MEMBER',payload:id});addToast(`Member ${name} removed`,'success');
    };
    return(
      <div>
        <div style={{display:'flex',justifyContent:'space-between',alignItems:'center',marginBottom:20,flexWrap:'wra
p',gap:12}}>
          <h2 className="font-orbitron glow-cyan" style={{fontSize:'1rem',letterSpacing:3,color:'var(--cyan)'}}>MEMBER
REGISTRY</h2>
          <div style={{display:'flex',gap:10,alignItems:'center'}}>
            <input className="input-field" value={search} onChange={e=>setSearch(e.target.value)} placeholder="ðŸ"⍰ Se
arch members..." style={{width:200}}/>
            <button className="btn-primary" onClick={()=>{setErrors({});setModal(true)}}>+ REGISTER</button>
          </div>
        </div>
        <div className="glass-static" style={{overflowX:'auto'}}>
          {sorted.length===0?<div className="empty-state"><div className="empty-icon">ðŸ‘¥</div>NO MEMBERS FOUND</div
>:(
            <table>
              <thead><tr>
                {[['membershipId','ID'],['name','NAME'],['memberType','TYPE'],['maxBooks','MAX']].map(([k,l])=>(
                  <th key={k} onClick={()=>sort(k)}>{l}<SortIcon k={k}/></th>
                ))}
                <th>ACTIVE LOANS</th><th>TOTAL FINES</th><th>ACTION</th>
              </tr></thead>
              <tbody>{sorted.map(m=>{
                const loans=state.loans.filter(l=>l.memberId===m.id&&l.status==='active').length;
                const fines=state.fines.filter(f=>f.memberId===m.id&&!f.paid).reduce((a,b)=>a+b.amount,0);
                return(
                  <tr key={m.id}>
                    <td><span className="badge badge-info">{m.membershipId}</span></td>
                    <td style={{fontWeight:600}}>{m.name}</td>
                    <td><span className="badge badge-grey">{m.memberType}</span></td>
                    <td style={{textAlign:'center'}}>{m.maxBooks}</td>
                    <td style={{textAlign:'center'}}>{loans}</td>
                    <td style={{color:fines>0?'#FF3860':'var(--success)',fontWeight:700}}>â‚¹{fines}</td>
                    <td><button className="btn-danger" style={{fontSize:'0.75rem',padding:'4px 10px'}} onClick={()=>ha
ndleDelete(m.id,m.name)}>REMOVE</button></td>
                  </tr>
                );
              })}</tbody>
            </table>
          )}
        </div>
```

```
        <Modal open={modal} onClose={()=>setModal(false)} title="REGISTER NEW MEMBER">
          {[['name','FULL NAME','text'],['username','USERNAME','text'],['password','PASSWORD','password'],['membership
Id','MEMBERSHIP ID','text'],['email','EMAIL','email']].map(([k,l,t])=>(
            <div key={k} style={{marginBottom:14}}>
              <label style={{display:'block',fontSize:'0.7rem',color:'rgba(0,255,255,0.6)',fontFamily:'Orbitron',lette
rSpacing:1,marginBottom:5}}>{l}</label>
              <input className="input-field" type={t} value={form[k]} onChange={e=>setForm(p=>({...p,[k]:e.target.valu
e}))}/>
              {errors[k]&&<p className="field-error">{errors[k]}</p>}
            </div>
          ))}
          <div style={{display:'grid',gridTemplateColumns:'1fr 1fr',gap:12,marginBottom:20}}>
            <div>
              <label style={{display:'block',fontSize:'0.7rem',color:'rgba(0,255,255,0.6)',fontFamily:'Orbitron',lette
rSpacing:1,marginBottom:5}}>TYPE</label>
              <select className="input-field" value={form.memberType} onChange={e=>setForm(p=>({...p,memberType:e.targ
et.value}))}>
                <option>Student</option><option>Faculty</option><option>Staff</option>
              </select>
            </div>
            <div>
              <label style={{display:'block',fontSize:'0.7rem',color:'rgba(0,255,255,0.6)',fontFamily:'Orbitron',lette
rSpacing:1,marginBottom:5}}>MAX BOOKS</label>
              <input className="input-field" type="number" min="1" max="10" value={form.maxBooks} onChange={e=>setForm
(p=>({...p,maxBooks:e.target.value}))}/>
            </div>
          </div>
          <button className="btn-primary" style={{width:'100%'}} onClick={handleSave}>REGISTER MEMBER</button>
        </Modal>
      </div>
    );
  }

  // â”€â”€â”€ RESERVATIONS (LIBRARIAN) â”€â”€â”€
  function LibReservations({state,dispatch,addToast}){
    const [search,setSearch]=useState('');
    const pending=state.reservations.filter(r=>r.status==='pending');
    const filtered=pending.filter(r=>
      r.memberName.toLowerCase().includes(search.toLowerCase())||
      r.bookTitle.toLowerCase().includes(search.toLowerCase())
    );
    const {sorted,sort,SortIcon}=useSortedData(filtered,'reservationDate');
    const cancel=(id)=>{dispatch({type:'CANCEL_RESERVATION',payload:id});addToast('Reservation cancelled','warning')};
    return(
      <div>
        <div style={{display:'flex',justifyContent:'space-between',alignItems:'center',marginBottom:20,flexWrap:'wra
p',gap:12}}>
          <h2 className="font-orbitron glow-cyan" style={{fontSize:'1rem',letterSpacing:3,color:'var(--cyan)'}}>RESERV
ATION QUEUE</h2>
          <input className="input-field" value={search} onChange={e=>setSearch(e.target.value)} placeholder="ðŸ”  Sear
ch..." style={{width:220}}/>
        </div>
        <div className="glass-static" style={{overflowX:'auto'}}>
          {sorted.length===0?<div className="empty-state"><div className="empty-icon">ðŸ“–</div>NO PENDING RESERVATION
S</div>:(
            <table>
              <thead><tr>
                {[['id','RES ID'],['memberName','MEMBER'],['bookTitle','BOOK'],['reservationDate','DATE'],['status','S
TATUS']].map(([k,l])=>(
                  <th key={k} onClick={()=>sort(k)}>{l}<SortIcon k={k}/></th>
                ))}
                <th>ACTION</th>
              </tr></thead>
              <tbody>{sorted.map(r=>(
                <tr key={r.id}>
                  <td><span className="badge badge-info">{r.id}</span></td>
                  <td>{r.memberName}</td>
                  <td style={{fontWeight:600}}>{r.bookTitle}</td>
                  <td>{formatDate(r.reservationDate)}</td>
                  <td><span className="badge badge-warning">PENDING</span></td>
                  <td><button className="btn-danger" style={{fontSize:'0.75rem',padding:'4px 10px'}} onClick={()=>canc
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el(r.id)}>CANCEL</button></td>
                </tr>
              ))}</tbody>
            </table>
          )}
        </div>
      </div>
    );
  }

  // â”€â”€â”€ REPORTS (LIBRARIAN) â”€â”€â”€â”€â”€
  function LibReports({state}){
    const [tab,setTab]=useState('all');
    const overdue=state.loans.filter(l=>{
      if(l.status!=='active')return false;
      return new Date(l.dueDate)<new Date();
    });
    const finesByMember=state.members.map(m=>({
      ...m,
      totalFines:state.fines.filter(f=>f.memberId===m.id).reduce((a,b)=>a+b.amount,0),
      paidFines:state.fines.filter(f=>f.memberId===m.id&&f.paid).reduce((a,b)=>a+b.amount,0),
    }));
    const tabs=['all','overdue','fines'];
    return(
      <div>
        <h2 className="font-orbitron glow-cyan" style={{fontSize:'1rem',letterSpacing:3,color:'var(--cyan)',marginBottom:20}}>SYSTEM REPORTS</h2>
        <div style={{display:'flex',gap:8,marginBottom:20,flexWrap:'wrap'}}>
          {tabs.map(t=>(
            <button key={t} onClick={()=>setTab(t)} className={tab===t?'btn-primary':'btn-sm'} style={{textTransform:'uppercase',letterSpacing:1}}>
              {t==='all'?'All Books':t==='overdue'?'Overdue Loans':'Fine Summary'}
            </button>
          ))}
        </div>
        <div className="glass-static" style={{overflowX:'auto'}}>
          {tab==='all'&&(state.books.length===0?<div className="empty-state"><div className="empty-icon">ðŸ“š</div>NO BOOKS</div>:(
            <table>
              <thead><tr><th>ISBN</th><th>TITLE</th><th>AUTHOR</th><th>CATEGORY</th><th>TOTAL</th><th>AVAILABLE</th></tr></thead>
              <tbody>{state.books.map(b=>(
                <tr key={b.isbn}><td>{b.isbn}</td><td>{b.title}</td><td>{b.author}</td><td>{b.category}</td><td>{b.totalCopies}</td>
                <td><span className={`badge ${b.availableCopies>0?'badge-success':'badge-danger'}`}>{b.availableCopies}</span></td></tr>
              ))}</tbody>
            </table>
          ))}
          {tab==='overdue'&&(overdue.length===0?<div className="empty-state"><div className="empty-icon">â°</div>NO OVERDUE LOANS</div>:(
            <table>
              <thead><tr><th>LOAN ID</th><th>MEMBER</th><th>BOOK</th><th>DUE DATE</th><th>DAYS OVERDUE</th><th>FINE (â‚¹)</th></tr></thead>
              <tbody>{overdue.map(l=>{
                const days=daysBetween(l.dueDate,today());
                return(<tr key={l.id} style={{background:'rgba(255,56,96,0.06)'}}>
                  <td>{l.id}</td><td>{l.memberName}</td><td>{l.bookTitle}</td>
                  <td style={{color:'#FF3860'}}>{formatDate(l.dueDate)}</td>
                  <td><span className="badge badge-danger">{days}d</span></td>
                  <td style={{color:'#FF3860',fontWeight:700}}>â‚¹{days*state.settings.fineRate}</td>
                </tr>);
              })}</tbody>
            </table>
          ))}
          {tab==='fines'&&(finesByMember.length===0?<div className="empty-state"><div className="empty-icon">ðŸ’°</div>NO DATA</div>:(
            <table>
              <thead><tr><th>MEMBER</th><th>TYPE</th><th>TOTAL FINES</th><th>PAID</th><th>UNPAID</th></tr></thead>
              <tbody>{finesByMember.map(m=>(
                <tr key={m.id}><td>{m.name}</td><td>{m.memberType}</td>
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                  <td style={{fontWeight:700}}>â‚¹{m.totalFines}</td>
                  <td style={{color:'var(--success)'}}>â‚¹{m.paidFines}</td>
                  <td style={{color:m.totalFines-m.paidFines>0?'#FF3860':'var(--success)'}}>â‚¹{m.totalFines-m.paidFines}</td>
                </tr>
              ))}</tbody>
            </table>
          ))}
        </div>
      </div>
    );
  }

  // â•â•â•
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    const canBorrow=currentUser.maxBooks-myLoans.length;
    return(
      <div>
        <div className="glass" style={{padding:24,marginBottom:24,cursor:'default'}}>
          <div style={{display:'flex',alignItems:'center',gap:16,flexWrap:'wrap'}}>
            <div style={{width:60,height:60,borderRadius:'50%',background:'linear-gradient(135deg,var(--cyan),var(--purple))',display:'flex',alignItems:'center',justifyContent:'center',fontSize:'1.8rem',boxShadow:'0 0 20px rgba(0,255,255,0.4)'}}>
              {currentUser.name.charAt(0)}
            </div>
            <div>
              <h2 className="font-orbitron" style={{fontSize:'1.1rem',color:'var(--cyan)',letterSpacing:2}}>WELCOME BACK, {currentUser.name.split(' ')[0].toUpperCase()}</h2>
              <p style={{fontFamily:'Rajdhani',color:'rgba(0,255,255,0.5)',fontSize:'0.85rem',letterSpacing:1}}>
                {currentUser.membershipId} â€¢ {currentUser.memberType} Member
              </p>
            </div>
          </div>
        </div>
        <div style={{display:'grid',gridTemplateColumns:'repeat(auto-fill,minmax(180px,1fr))',gap:16}}>
          <StatCard icon="ðŸ“–" label="BOOKS BORROWED" value={myLoans.length} color="var(--cyan)"/>
          <StatCard icon="âœ…" label="CAN BORROW" value={canBorrow} color="var(--success)"/>
          <StatCard icon="ðŸ’³" label="PENDING FINES (â‚¹)" value={myFines} color={myFines>0?'var(--orange)':'var(--success)'} pulse={myFines>0}/>
        </div>
      </div>
    );
  }

  function MemSearch({currentUser,state,dispatch,addToast}){
    const [search,setSearch]=useState('');
    const filtered=state.books.filter(b=>
      b.title.toLowerCase().includes(search.toLowerCase())||
      b.author.toLowerCase().includes(search.toLowerCase())||
      b.category.toLowerCase().includes(search.toLowerCase())
    );
    const myActiveLoans=state.loans.filter(l=>l.memberId===currentUser.id&&l.status==='active');
    const handleReserve=(book)=>{
      if(!state.settings.enableReservations){addToast('Reservations are currently disabled','error');return}
      const existing=state.reservations.find(r=>r.memberId===currentUser.id&&r.bookIsbn===book.isbn&&r.status==='pending');
      if(existing){addToast('You already have a reservation for this book','warning');return}
      dispatch({type:'ADD_RESERVATION',payload:{memberId:currentUser.id,memberName:currentUser.name,bookIsbn:book.isbn,bookTitle:book.title,reservationDate:today()}});
      addToast(`Reserved "${book.title}"`,'success');
    };
    const handleRequestIssue=(book)=>{
      const memberLoans=myActiveLoans.length;
      if(memberLoans>=currentUser.maxBooks){addToast(`Borrow limit reached (${currentUser.maxBooks} books max)`,'error');return}
      const alreadyBorrowed=myActiveLoans.find(l=>l.bookIsbn===book.isbn);
      if(alreadyBorrowed){addToast('You already have this book borrowed','warning');return}
      dispatch({type:'ISSUE_BOOK',payload:{memberId:currentUser.id,bookIsbn:book.isbn,issueDate:today(),dueDate:addDays(today(),15),memberName:currentUser.name,bookTitle:book.title}});
      addToast(`"${book.title}" issued to you! Due: ${formatDate(addDays(today(),15))}`,'success');
    };
    return(
      <div>
        <h2 className="font-orbitron glow-cyan" style={{fontSize:'1rem',letterSpacing:3,color:'var(--cyan)',marginBottom:20}}>SEARCH BOOKS</h2>
        <input className="input-field" value={search} onChange={e=>setSearch(e.target.value)} placeholder="ðŸ”  Search by title, author, or category..." style={{marginBottom:24}}/>
        {filtered.length===0?<div className="empty-state"><div className="empty-icon">ðŸ”</div>NO BOOKS MATCH YOUR SEARCH</div>:(
          <div style={{display:'grid',gridTemplateColumns:'repeat(auto-fill,minmax(280px,1fr))',gap:16}}>
            {filtered.map(b=>(
              <div key={b.isbn} className="glass" style={{padding:20,cursor:'default'}}>
                <div style={{display:'flex',justifyContent:'space-between',alignItems:'flex-start',marginBottom:10}}>
                  <span className="badge badge-grey" style={{fontSize:'0.65rem'}}>{b.category}</span>
                  <span className={`badge ${b.availableCopies>0?'badge-success':'badge-danger'}`}>{b.availableCopies>0?`${b.availableCopies} AVAIL`:'NONE'}</span>
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                </div>
                <h3 style={{fontFamily:'Orbitron',fontSize:'0.85rem',color:'var(--cyan)',marginBottom:6,letterSpacing:1}}>{b.title}</h3>
                <p style={{fontFamily:'Rajdhani',fontSize:'0.9rem',color:'rgba(224,247,250,0.7)',marginBottom:14}}>{b.author}</p>
                <div style={{display:'flex',gap:8}}>
                  {b.availableCopies>0?(
                    <button className="btn-sm" style={{flex:1}} onClick={()=>handleRequestIssue(b)}>BORROW</button>
                  ):(
                    <button className="btn-warning" style={{flex:1}} onClick={()=>handleReserve(b)}>RESERVE</button>
                  )}
                </div>
              </div>
            ))}
          </div>
        )}
      </div>
    );
  }

  function MemLoans({currentUser,state}){
    const [search,setSearch]=useState('');
    const myLoans=state.loans.filter(l=>l.memberId===currentUser.id);
    const filtered=myLoans.filter(l=>l.bookTitle.toLowerCase().includes(search.toLowerCase()));
    const {sorted,sort,SortIcon}=useSortedData(filtered,'issueDate');
    return(
      <div>
        <div style={{display:'flex',justifyContent:'space-between',alignItems:'center',marginBottom:20,flexWrap:'wrap',gap:12}}>
          <h2 className="font-orbitron glow-cyan" style={{fontSize:'1rem',letterSpacing:3,color:'var(--cyan)'}}>MY LOANS</h2>
          <input className="input-field" value={search} onChange={e=>setSearch(e.target.value)} placeholder="ðŸ”  Search..." style={{width:220}}/>
        </div>
        <div className="glass-static" style={{overflowX:'auto'}}>
          {sorted.length===0?<div className="empty-state"><div className="empty-icon">ðŸ“–</div>NO LOANS YET</div>:(
            <table>
              <thead><tr>
                {[['id','LOAN ID'],['bookTitle','BOOK'],['issueDate','ISSUED'],['dueDate','DUE']].map(([k,l])=>(
                  <th key={k} onClick={()=>sort(k)}>{l}<SortIcon k={k}/></th>
                ))}
                <th>STATUS</th><th>FINE</th>
              </tr></thead>
              <tbody>{sorted.map(l=>{
                const today_=today();
                const overdueDays=l.status==='active'?daysBetween(l.dueDate,today_):0;
                const isOverdue=overdueDays>0&&l.status==='active';
                const relatedFine=state.fines.find(f=>f.loanId===l.id);
                return(
                  <tr key={l.id}>
                    <td><span className="badge badge-info">{l.id}</span></td>
                    <td style={{fontWeight:600}}>{l.bookTitle}</td>
                    <td>{formatDate(l.issueDate)}</td>
                    <td style={{color:isOverdue?'#FF3860':'var(--text)'}}>{formatDate(l.dueDate)}</td>
                    <td>
                      {l.status==='returned'?<span className="badge badge-grey">RETURNED</span>:
                       isOverdue?<span className="badge badge-danger">OVERDUE</span>:
                       <span className="badge badge-success">ACTIVE</span>}
                    </td>
                    <td style={{color:relatedFine?'#FF3860':'var(--success)',fontWeight:700}}>
                      {relatedFine?`â‚¹${relatedFine.amount}`:'â‚¹0'}
                    </td>
                  </tr>
                );
              })}</tbody>
            </table>
          )}
        </div>
      </div>
    );
```

```
  }

  function MemReservations({currentUser,state,dispatch,addToast}){
    const [search,setSearch]=useState('');
    const myRes=state.reservations.filter(r=>r.memberId===currentUser.id);
    const filtered=myRes.filter(r=>r.bookTitle.toLowerCase().includes(search.toLowerCase()));
    const {sorted,sort,SortIcon}=useSortedData(filtered,'reservationDate');
    const cancel=(id)=>{dispatch({type:'CANCEL_RESERVATION',payload:id});addToast('Reservation cancelled','warning')};
    return(
      <div>
        <div style={{display:'flex',justifyContent:'space-between',alignItems:'center',marginBottom:20,flexWrap:'wrap',gap:12}}>
          <h2 className="font-orbitron glow-cyan" style={{fontSize:'1rem',letterSpacing:3,color:'var(--cyan)'}}>MY RESERVATIONS</h2>
          <input className="input-field" value={search} onChange={e=>setSearch(e.target.value)} placeholder="ðŸ” Search..." style={{width:220}}/>
        </div>
        <div className="glass-static" style={{overflowX:'auto'}}>
          {sorted.length===0?<div className="empty-state"><div className="empty-icon">ðŸ“–</div>NO RESERVATIONS</div>:(
            <table>
              <thead><tr>
                {[['id','RES ID'],['bookTitle','BOOK'],['reservationDate','DATE'],['status','STATUS']].map(([k,l])=>(
                  <th key={k} onClick={()=>sort(k)}>{l}<SortIcon k={k}/></th>
                ))}
                <th>ACTION</th>
              </tr></thead>
              <tbody>{sorted.map(r=>(
                <tr key={r.id}>
                  <td><span className="badge badge-info">{r.id}</span></td>
                  <td style={{fontWeight:600}}>{r.bookTitle}</td>
                  <td>{formatDate(r.reservationDate)}</td>
                  <td><span className={`badge ${r.status==='pending'?'badge-warning':'badge-grey'}`}>{r.status.toUpperCase()}</span></td>
                  <td>
                    {r.status==='pending'&&<button className="btn-danger" style={{fontSize:'0.75rem',padding:'4px 10px'}} onClick={()=>cancel(r.id)}>CANCEL</button>}
                  </td>
                </tr>
              ))}</tbody>
            </table>
          )}
        </div>
      </div>
    );
  }

  function MemFines({currentUser,state,dispatch,addToast}){
    const [paidId,setPaidId]=useState(null);
    const [search,setSearch]=useState('');
    const myFines=state.fines.filter(f=>f.memberId===currentUser.id);
    const filtered=myFines.filter(f=>String(f.id).toLowerCase().includes(search.toLowerCase())||f.bookTitle?.toLowerCase().includes(search.toLowerCase()));
    const {sorted,sort,SortIcon}=useSortedData(filtered,'id');
    const payFine=(id)=>{
      setPaidId(id);
      setTimeout(()=>{dispatch({type:'PAY_FINE',payload:id});addToast('Fine paid successfully! âœ¨','success');setPaidId(null)},800);
    };
    const unpaid=myFines.filter(f=>!f.paid).reduce((a,b)=>a+b.amount,0);
    return(
      <div>
        <div style={{display:'flex',justifyContent:'space-between',alignItems:'center',marginBottom:20,flexWrap:'wrap',gap:12}}>
          <h2 className="font-orbitron glow-cyan" style={{fontSize:'1rem',letterSpacing:3,color:'var(--cyan)'}}>MY FINES</h2>
          <input className="input-field" value={search} onChange={e=>setSearch(e.target.value)} placeholder="ðŸ” Search..." style={{width:220}}/>
        </div>
        {unpaid>0&&(
          <div className="glass" style={{padding:16,marginBottom:20,cursor:'default',borderColor:'rgba(255,107,0,0.
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5)',boxShadow:'0 0 20px rgba(255,107,0,0.2)'}}>
            <p style={{fontFamily:'Orbitron',fontSize:'0.85rem',color:'#FF6B00'}}>âš ï¸ TOTAL UNPAID: â‚¹{unpaid}</p>
          </div>
        )}
        <div className="glass-static" style={{overflowX:'auto'}}>
          {sorted.length===0?<div className="empty-state"><div className="empty-icon">âœ…</div>NO FINES â€” EXCELLENT RECORD!</div>:(
            <table>
              <thead><tr>
                {[['id','FINE ID'],['bookTitle','BOOK'],['amount','AMOUNT'],['reason','REASON']].map(([k,l])=>(
                  <th key={k} onClick={()=>sort(k)}>{l}<SortIcon k={k}/></th>
                ))}
                <th>STATUS</th><th>ACTION</th>
              </tr></thead>
              <tbody>{sorted.map(f=>(
                <tr key={f.id} style={paidId===f.id?{background:'rgba(0,255,136,0.1)',transition:'background 0.5s'}:{}}>
                  <td><span className="badge badge-warning">{f.id}</span></td>
                  <td style={{fontWeight:600}}>{f.bookTitle||'â€”'}</td>
                  <td style={{color:f.paid?'var(--success)':'#FF3860',fontWeight:700}}>â‚¹{f.amount}</td>
                  <td>{f.reason||'Overdue'}</td>
                  <td><span className={`badge ${f.paid?'badge-success':'badge-danger'}`}>{f.paid?'PAID':'UNPAID'}</span></td>
                  <td>
                    {!f.paid&&<button className="btn-success" onClick={()=>payFine(f.id)} style={{fontSize:'0.78rem'}}>PAY â‚¹{f.amount}</button>}
                    {f.paid&&<span style={{color:'var(--success)',fontSize:'1.2rem'}}>âœ“</span>}
                  </td>
                </tr>
              ))}</tbody>
            </table>
          )}
        </div>
      </div>
    );
  }

  function MemProfile({currentUser,state,dispatch,addToast}){
    const [editing,setEditing]=useState(false);
    const [form,setForm]=useState({email:currentUser.email||'',password:''});
    const [errors,setErrors]=useState({});
    const save=()=>{
      const e={};
      if(!form.email.trim()||!form.email.includes('@'))e.email='Valid email required';
      if(form.password&&form.password.length<4)e.password='Min 4 characters';
      setErrors(e);if(Object.keys(e).length)return;
      dispatch({type:'UPDATE_MEMBER_PROFILE',payload:{id:currentUser.id,email:form.email,password:form.password||currentUser.password}});
      addToast('Profile updated successfully','success');setEditing(false);
    };
    const myLoans=state.loans.filter(l=>l.memberId===currentUser.id);
    return(
      <div>
        <h2 className="font-orbitron glow-cyan" style={{fontSize:'1rem',letterSpacing:3,color:'var(--cyan)',marginBottom:24}}>MY PROFILE</h2>
        <div style={{display:'grid',gridTemplateColumns:'repeat(auto-fit,minmax(300px,1fr))',gap:20}}>
          <div className="glass" style={{padding:24,cursor:'default'}}>
            <div style={{display:'flex',justifyContent:'space-between',alignItems:'center',marginBottom:20}}>
              <p className="font-orbitron" style={{fontSize:'0.72rem',color:'var(--cyan)',letterSpacing:2}}>IDENTITY</p>
              <button className="btn-sm" onClick={()=>setEditing(!editing)}>{editing?'CANCEL':'EDIT'}</button>
            </div>
            {[
              ['Name',currentUser.name],
              ['Username',currentUser.username],
              ['Membership ID',currentUser.membershipId],
              ['Type',currentUser.memberType],
              ['Max Books',currentUser.maxBooks],
            ].map(([k,v])=>(
              <div key={k} style={{marginBottom:14,paddingBottom:14,borderBottom:'1px solid rgba(0,255,255,0.08)'}}>
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                <p style={{fontSize:'0.7rem',color:'rgba(0,255,255,0.5)',fontFamily:'Orbitron',letterSpacing:1,marginBottom:4}}>{k.toUpperCase()}</p>
                <p style={{fontFamily:'Rajdhani',fontSize:'1rem',fontWeight:600}}>{v}</p>
              </div>
            ))}
            {editing&&(
              <div>
                <div style={{marginBottom:14}}>
                  <label style={{display:'block',fontSize:'0.7rem',color:'rgba(0,255,255,0.6)',fontFamily:'Orbitron',letterSpacing:1,marginBottom:5}}>EMAIL</label>
                  <input className="input-field" value={form.email} onChange={e=>setForm(p=>({...p,email:e.target.value}))}/>
                  {errors.email&&<p className="field-error">{errors.email}</p>}
                </div>
                <div style={{marginBottom:16}}>
                  <label style={{display:'block',fontSize:'0.7rem',color:'rgba(0,255,255,0.6)',fontFamily:'Orbitron',letterSpacing:1,marginBottom:5}}>NEW PASSWORD (LEAVE BLANK TO KEEP)</label>
                  <input className="input-field" type="password" value={form.password} onChange={e=>setForm(p=>({...p,password:e.target.value}))}/>
                  {errors.password&&<p className="field-error">{errors.password}</p>}
                </div>
                <button className="btn-primary" style={{width:'100%'}} onClick={save}>SAVE CHANGES</button>
              </div>
            )}
          </div>
          <div className="glass" style={{padding:24,cursor:'default'}}>
            <p className="font-orbitron" style={{fontSize:'0.72rem',color:'var(--cyan)',letterSpacing:2,marginBottom:20}}>STATISTICS</p>
            {[
              ['Total Borrowed',myLoans.length,'ðŸ“š'],
              ['Currently Active',myLoans.filter(l=>l.status==='active').length,'ðŸ“–'],
              ['Returned',myLoans.filter(l=>l.status==='returned').length,'âœ…'],
              ['Total Fines',`â‚¹${state.fines.filter(f=>f.memberId===currentUser.id).reduce((a,b)=>a+b.amount,0)}`,'ðŸ’°'],
            ].map(([k,v,i])=>(
              <div key={k} style={{display:'flex',justifyContent:'space-between',alignItems:'center',padding:'12px 0',borderBottom:'1px solid rgba(0,255,255,0.08)'}}>
                <div style={{display:'flex',alignItems:'center',gap:10}}>
                  <span>{i}</span>
                  <span style={{fontFamily:'Rajdhani',fontSize:'0.9rem',color:'rgba(224,247,250,0.7)'}}>{k}</span>
                </div>
                <span style={{fontFamily:'Orbitron',fontSize:'1rem',color:'var(--cyan)',fontWeight:700}}>{v}</span>
              </div>
            ))}
          </div>
        </div>
      </div>
    );
  }

  // â•â•â•
```

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            <p className="font-orbitron" style={{fontSize:'0.65rem',color:'var(--cyan)',letterSpacing:2,opacity:0.7}}>ADMIN PANEL</p>
          </div>
          {navItems.map(n=>(
            <div key={n.id} className={`nav-item ${page===n.id?'nav-active':''}`} onClick={()=>setPage(n.id)}>
              <span className="nav-icon">{n.icon}</span>
              <span className="nav-label" style={{fontFamily:'Rajdhani',fontWeight:600}}>{n.label}</span>
            </div>
          ))}
        </nav>
        <div className="main-area" style={{flex:1,overflowY:'auto',padding:'24px'}}>
          <div className="page-fade" key={page}>{pageComp[page]}</div>
        </div>
        <div className="bottom-bar">
          {navItems.map(n=>(
            <div key={n.id} onClick={()=>setPage(n.id)} style={{display:'flex',flexDirection:'column',alignItems:'center',gap:3,cursor:'pointer',padding:'4px 8px',opacity:page===n.id?1:0.5}}>
              <span style={{fontSize:'1.1rem'}}>{n.icon}</span>
              <span style={{fontSize:'0.55rem',fontFamily:'Orbitron',color:page===n.id?'var(--cyan)':'var(--text)',letterSpacing:0.5}}>{n.label.split(' ')[0]}</span>
            </div>
          ))}
        </div>
      </div>
    );
  }

  function AdminUsers({state,dispatch,addToast}){
    const [tab,setTab]=useState('librarians');
    const [modal,setModal]=useState(false);
    const [form,setForm]=useState({name:'',username:'',password:'',role:'librarian'});
    const [errors,setErrors]=useState({});
    const [search,setSearch]=useState('');
    const allUsers=[...INITIAL_USERS,...state.members].filter(u=>u.role!=='admin');
    const librarians=allUsers.filter(u=>u.role==='librarian');
    const members_=state.members;
    const list=tab==='librarians'?librarians:members_;
    const filtered=list.filter(u=>u.name.toLowerCase().includes(search.toLowerCase())||u.username?.toLowerCase().includes(search.toLowerCase()));
    const {sorted,sort,SortIcon}=useSortedData(filtered,'name');
    const validate=()=>{
      const e={};
      if(!form.name.trim())e.name='Name required';
      if(!form.username.trim())e.username='Username required';
      if(allUsers.find(u=>u.username===form.username))e.username='Username taken';
      if(!form.password||form.password.length<4)e.password='Min 4 characters';
      setErrors(e);return Object.keys(e).length===0;
    };
    const handleCreate=()=>{
      if(!validate())return;
      if(form.role==='member'){
        addToast('Use Librarian panel to register members','warning');return;
      }
      addToast(`Account created for ${form.name} (${form.role})`,'success');
      setModal(false);
    };
    const handleDelete=(u)=>{
      if(u.role==='librarian'&&INITIAL_USERS.find(x=>x.username===u.username)){addToast('Cannot delete default accounts','error');return}
      dispatch({type:'DELETE_MEMBER',payload:u.id});addToast(`${u.name} account removed`,'success');
    };
    return(
      <div>
        <div style={{display:'flex',justifyContent:'space-between',alignItems:'center',marginBottom:20,flexWrap:'wrap',gap:12}}>
          <h2 className="font-orbitron glow-cyan" style={{fontSize:'1rem',letterSpacing:3,color:'var(--cyan)'}}>USER MANAGEMENT</h2>
          <div style={{display:'flex',gap:10,alignItems:'center'}}>
            <input className="input-field" value={search} onChange={e=>setSearch(e.target.value)} placeholder="ðŸ”  Search users..." style={{width:200}}/>
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            <button className="btn-primary" onClick={()=>{setErrors({});setModal(true)}}>+ CREATE</button>
          </div>
        </div>
        <div style={{display:'flex',gap:8,marginBottom:16}}>
          {[['librarians','Librarians'],['members','Members']].map(([k,l])=>(
            <button key={k} onClick={()=>setTab(k)} className={tab===k?'btn-primary':'btn-sm'} style={{letterSpacing:1}}>{l.toUpperCase()}</button>
          ))}
        </div>
        <div className="glass-static" style={{overflowX:'auto'}}>
          {sorted.length===0?<div className="empty-state"><div className="empty-icon">ðŸ‘¤</div>NO USERS</div>:(
            <table>
              <thead><tr>
                <th onClick={()=>sort('name')}>NAME <SortIcon k="name"/></th>
                <th onClick={()=>sort('username')}>USERNAME <SortIcon k="username"/></th>
                <th>ROLE</th>
                {tab==='members'&&<><th>TYPE</th><th>MEMBERSHIP ID</th></>}
                <th>ACTION</th>
              </tr></thead>
              <tbody>{sorted.map(u=>(
                <tr key={u.id||u.username}>
                  <td style={{fontWeight:600}}>{u.name}</td>
                  <td style={{fontFamily:'Orbitron',fontSize:'0.75rem',color:'rgba(0,255,255,0.7)'}}>{u.username}</td>
                  <td><span className={`badge ${u.role==='librarian'?'badge-info':'badge-success'}`}>{u.role?.toUpperCase()}</span></td>
                  {tab==='members'&&<><td>{u.memberType}</td><td>{u.membershipId}</td></>}
                  <td>
                    {!(INITIAL_USERS.find(x=>x.username===u.username))&&(
                      <button className="btn-danger" style={{fontSize:'0.75rem',padding:'4px 10px'}} onClick={()=>handleDelete(u)}>DELETE</button>
                    )}
                  </td>
                </tr>
              ))}</tbody>
            </table>
          )}
        </div>
        <Modal open={modal} onClose={()=>setModal(false)} title="CREATE ACCOUNT">
          {[['name','FULL NAME','text'],['username','USERNAME','text'],['password','PASSWORD','password']].map(([k,l,t])=>(
            <div key={k} style={{marginBottom:14}}>
              <label style={{display:'block',fontSize:'0.7rem',color:'rgba(0,255,255,0.6)',fontFamily:'Orbitron',letterSpacing:1,marginBottom:5}}>{l}</label>
              <input className="input-field" type={t} value={form[k]} onChange={e=>setForm(p=>({...p,[k]:e.target.value}))}/>
              {errors[k]&&<p className="field-error">{errors[k]}</p>}
            </div>
          ))}
          <div style={{marginBottom:20}}>
            <label style={{display:'block',fontSize:'0.7rem',color:'rgba(0,255,255,0.6)',fontFamily:'Orbitron',letterSpacing:1,marginBottom:5}}>ROLE</label>
            <select className="input-field" value={form.role} onChange={e=>setForm(p=>({...p,role:e.target.value}))}>
              <option value="librarian">Librarian</option>
            </select>
          </div>
          <button className="btn-primary" style={{width:'100%'}} onClick={handleCreate}>CREATE ACCOUNT</button>
        </Modal>
      </div>
    );
  }

  function Toggle({value,onChange,label}){
    return(
      <div style={{display:'flex',alignItems:'center',justifyContent:'space-between',padding:'16px 0',borderBottom:'1px solid rgba(0,255,255,0.08)'}}>
        <span style={{fontFamily:'Rajdhani',fontSize:'1rem',fontWeight:600}}>{label}</span>
        <div className={`toggle-track ${value?'on':''}`} onClick={()=>onChange(!value)}>
          <div className="toggle-thumb"/>
        </div>
      </div>
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    );
  }

  function AdminSettings({state,dispatch,addToast}){
    const s=state.settings;
    const update=(k,v)=>{dispatch({type:'UPDATE_SETTING',payload:{[k]:v}});addToast('Setting updated','success')};
    return(
      <div>
        <h2 className="font-orbitron glow-cyan" style={{fontSize:'1rem',letterSpacing:3,color:'var(--cyan)',marginBottom:24}}>SYSTEM SETTINGS</h2>
        <div className="glass-static" style={{padding:28,maxWidth:500}}>
          <Toggle value={s.enableReservations} onChange={v=>update('enableReservations',v)} label="Enable Reservations"/>
          <Toggle value={s.enableFineAutoCalc} onChange={v=>update('enableFineAutoCalc',v)} label="Enable Fine Auto-Calculate"/>
          <div style={{display:'flex',alignItems:'center',justifyContent:'space-between',padding:'16px 0',borderBottom:'1px solid rgba(0,255,255,0.08)'}}>
            <span style={{fontFamily:'Rajdhani',fontSize:'1rem',fontWeight:600}}>Max Books Per Member</span>
            <input className="input-field" type="number" min="1" max="20" value={s.maxBooksDefault} onChange={e=>update('maxBooksDefault',parseInt(e.target.value))} style={{width:80}}/>
          </div>
          <div style={{display:'flex',alignItems:'center',justifyContent:'space-between',padding:'16px 0'}}>
            <span style={{fontFamily:'Rajdhani',fontSize:'1rem',fontWeight:600}}>Fine Rate Per Day (â‚¹)</span>
            <input className="input-field" type="number" min="1" max="50" value={s.fineRate} onChange={e=>update('fineRate',parseInt(e.target.value))} style={{width:80}}/>
          </div>
        </div>
      </div>
    );
  }

  function AdminReports({state}){
    const totalBooks=state.books.reduce((a,b)=>a+b.totalCopies,0);
    const availBooks=state.books.reduce((a,b)=>a+b.availableCopies,0);
    const activeLoans=state.loans.filter(l=>l.status==='active').length;
    const overdueLoans=state.loans.filter(l=>l.status==='active'&&new Date(l.dueDate)<new Date()).length;
    const totalFines=state.fines.reduce((a,b)=>a+b.amount,0);
    const paidFines=state.fines.filter(f=>f.paid).reduce((a,b)=>a+b.amount,0);
    return(
      <div>
        <h2 className="font-orbitron glow-cyan" style={{fontSize:'1rem',letterSpacing:3,color:'var(--cyan)',marginBottom:24}}>SYSTEM REPORT</h2>
        <div style={{display:'grid',gridTemplateColumns:'repeat(auto-fill,minmax(160px,1fr))',gap:14,marginBottom:28}}>
          <StatCard icon="ðŸ“š" label="TOTAL BOOKS" value={totalBooks} color="var(--cyan)"/>
          <StatCard icon="âœ…" label="AVAILABLE" value={availBooks} color="var(--success)"/>
          <StatCard icon="ðŸ“¤" label="ACTIVE LOANS" value={activeLoans} color="var(--cyan)"/>
          <StatCard icon="âš ï¸" label="OVERDUE" value={overdueLoans} color="var(--orange)" pulse/>
          <StatCard icon="ðŸ‘¥" label="MEMBERS" value={state.members.length} color="var(--purple)"/>
          <StatCard icon="ðŸ’°" label="TOTAL FINES (â‚¹)" value={totalFines} color="var(--orange)"/>
          <StatCard icon="âœ”ï¸" label="PAID (â‚¹)" value={paidFines} color="var(--success)"/>
          <StatCard icon="â" label="UNPAID (â‚¹)" value={totalFines-paidFines} color="var(--danger)"/>
        </div>
        <div className="glass-static" style={{overflowX:'auto'}}>
          <table>
            <thead><tr><th>METRIC</th><th>VALUE</th></tr></thead>
            <tbody>
              {[
                ['Total Book Titles',state.books.length],
                ['Total Copies',totalBooks],
                ['Available Copies',availBooks],
                ['Checked Out Copies',totalBooks-availBooks],
                ['Active Loans',activeLoans],
                ['Overdue Loans',overdueLoans],
                ['Total Reservations',state.reservations.length],
                ['Pending Reservations',state.reservations.filter(r=>r.status==='pending').length],
                ['Total Members',state.members.length],
                ['Total Fines (â‚¹)',totalFines],
                ['Paid Fines (â‚¹)',paidFines],
                ['Unpaid Fines (â‚¹)',totalFines-paidFines],
```

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              ].map(([k,v])=>(
                <tr key={k}><td>{k}</td><td style={{fontFamily:'Orbitron',color:'var(--cyan)',fontWeight:700}}>{v}</td></tr>
              ))}
            </tbody>
          </table>
        </div>
      </div>
    );
  }

  // â•â•â•
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    // â•â•â•
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